



UIT UNIVERSITY

PAKISTAN ELECTRICITY SMART BILL PREDICTOR & OPTIMIZATION DASHBOARD

Project Report (Artificial Intelligence)

Course: Artificial Intelligence

Instructor: Miss Maham

Submission Date: 17-APR-2026

Group Members

- ❖ Muhammad Bilal (24FA-050-SE)
 - ❖ Fahad Ahmed (24FA-055-SE)
 - ❖ Sarim Bin Tahir (24FA-012-SE)
 - ❖ Muhammad Daniyal (24FA-048-SE)
-

GitHub Repository Link : <https://github.com/FahadAhmed-code/python-project.git>

Abstract

- AI/ML-powered dashboard for electricity bill prediction and optimization tailored for Pakistan
- Implements three predictive models: Random Forest, XGBoost, and Facebook Prophet
- Uses K-Means clustering with PCA visualization to segment consumers into four behavioral groups
- Integrates NEPRA-compliant billing engine for accurate PKR bill computation across tariff slabs
- Agentic AI recommendations generated via Groq LLM (RAG-based prompt engineering)
- Interactive three-tab Streamlit dashboard with dark mode, real-time sliders, and animated charts
- Analyzes 50,000 real household power consumption records with seasonal decomposition and EDA
- Demonstrates practical AI application to Pakistan's energy sustainability challenge

1. Introduction

Pakistan faces a severe electricity crisis characterized by unpredictable billing, high tariff slabs, and limited consumer awareness about energy optimization. Traditional billing systems provide no forward-looking insights, leaving households unable to plan or reduce their consumption strategically.

This project develops a Pakistan Electricity Smart Bill Predictor & Optimization Dashboard — an end-to-end AI/ML system that transforms raw household power data into actionable predictions and personalized energy-saving recommendations.

The system models electricity consumption patterns using supervised machine learning, segments consumers via unsupervised clustering, calculates bills using the official NEPRA tariff structure, and provides LLM-generated advisory using prescriptive AI. The Streamlit dashboard makes these capabilities accessible through an intuitive, visually rich interface.

2. Objectives

- Implement and compare supervised regression models (Random Forest, XGBoost, Prophet) for consumption forecasting
- Design a complete data engineering pipeline with outlier removal, missing-value imputation, and feature extraction
- Segment consumers using K-Means clustering and visualize groups using PCA dimensionality reduction
- Build a fully compliant NEPRA billing engine supporting all five tariff slabs and 17% tax calculation
- Integrate an LLM-powered advisory module providing personalized energy-saving recommendations with PKR savings estimates

- Deliver an interactive, dark-mode Streamlit dashboard with real-time what-if simulation capabilities
- Analyze 50,000 real-world household power consumption records for pattern discovery

3. System Overview

The system represents a comprehensive AI pipeline where raw electricity consumption data flows through preprocessing, modeling, segmentation, billing, and advisory stages — all unified in an interactive dashboard.

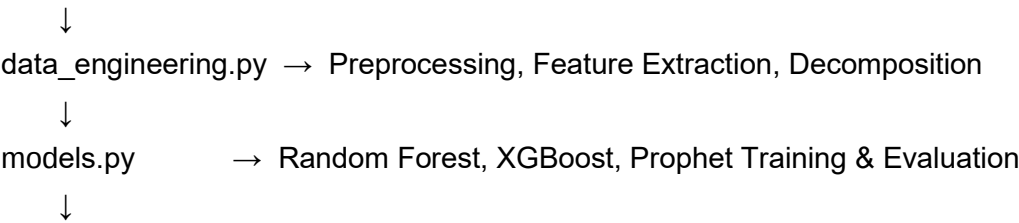
Concept Mapping

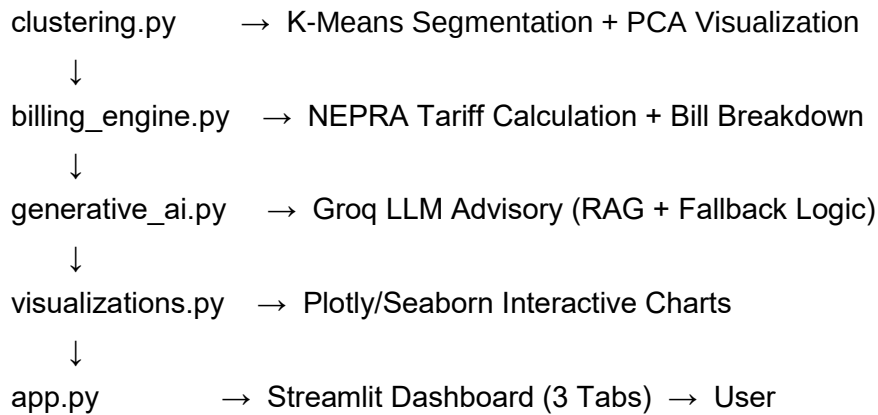
Real-World Concept	System Representation
Pakistan Household	50,000-row household_power_consumption_50k.txt dataset
Electricity Consumption	Global_active_power (kWh) — primary prediction target
Monthly Bill	NEPRA tariff calculation: $\text{Base} + (\text{Units} \times \text{Rate}) + 17\% \text{ Tax}$
Consumer Type	K-Means cluster: Protected / Middle-Class / Commercial / Industrial
Seasonal Pattern	Sinusoidal temperature model + temporal feature extraction
AI Advisor	Groq LLM (Mixtral-8x7b) with RAG-based tariff-guided prompts
Dashboard	3-tab Streamlit app with dark theme (#0e1117) and neon accents

4. System Architecture

The system follows a modular pipeline architecture where each component handles a clearly defined responsibility:

Raw Data (household_power_consumption_50k.txt)





- Modular design ensures each file can be independently developed, tested, and extended
- app.py orchestrates all components and serves as the single entry point
- config.py centralizes all hyperparameters, tariff slabs, and theme constants

5. Workflow

Load Dataset (household_power_consumption_50k.txt or synthetic fallback)



Handle Missing Data (Seasonal Median Imputation)



Detect & Remove Outliers (IQR Method, 1.5× multiplier)



Extract Temporal Features (month, quarter, season, peak hours)



User Selects: Dashboard / AI Analytics / AI Bill Optimizer Tab



Tab 1: Live Sliders → Bill Prediction → Breakdown Pie Chart

Tab 2: EDA / Model Training / Clustering / Feature Analysis

Tab 3: User Profile Input → LLM Recommendations → Action Plan



Display Metrics, Charts, and Best Model Highlight

- Every slider interaction triggers a real-time bill recalculation via the BillingEngine
- Model training uses an 80/20 train-test split with fixed random_state=42 for reproducibility
- LLM fallback logic ensures recommendations are always provided even without an API key

6. Modules & Algorithms Used

Module A — Data Engineering (data_engineering.py)

Function	Technique	Purpose
handle_missing_data()	Seasonal Median Imputation	Preserve seasonal patterns instead of dropping rows
detect_outliers()	IQR Method (1.5× multiplier)	Remove anomalous readings while keeping seasonal variance
extract_temporal_features()	Pandas datetime extraction	Generate month, quarter, season, peak/off-peak flags
decompose_timeseries()	Statsmodels seasonal_decompose	Split signal into Trend + Seasonality + Residual
prepare_ml_data()	MinMax / Z-Score normalization	Scale features for ML model training

Module B — Predictive Modeling (models.py)

Algorithm	Type	Key Property	Best Use Case
Random Forest	Supervised Regression	Ensemble of 100 decision trees	Robust baseline, handles non-linearity
XGBoost	Gradient Boosting	Sequential error correction, depth=6	High accuracy with structured data
Facebook Prophet	Time-Series Forecast	Handles seasonality + trend + holidays	3-month electricity demand forecasting

Evaluation Metrics

Metric	Formula	Interpretation
R ² Score	$1 - (SS_{res} / SS_{tot})$	Proportion of variance explained (0–1, higher is better)
RMSE	$\sqrt{(\sum (y_{actual} - y_{pred})^2 / n)}$	Average prediction error in kWh (lower is better)
MAE	$\sum y_{actual} - y_{pred} / n$	Mean absolute error in kWh (lower is better)
MAPE	$(100/n) \times \sum y_{actual} - y_{pred} / y_{actual}$	Percentage error for interpretability

Module C — Consumer Segmentation (clustering.py)

K-Means clustering (k=4) groups consumers by electricity behaviour using three features: consumption (kWh), temperature (°C), and humidity (%). PCA reduces dimensionality to 2 components for 2D scatter visualization.

Cluster	Segment Name	Consumption Range	Behaviour
Cluster 0	Low-Income / Protected	0 – 100 kWh	Minimal usage, protected tariff
Cluster 1	Middle-Class / Slab-Breacher	101 – 300 kWh	Moderate usage, at risk of slab jump
Cluster 2	Heavy Commercial / AC-Heavy	301 – 500 kWh	High usage, significant optimization potential
Cluster 3	Industrial / Extreme	500+ kWh	Industrial load, highest tariff rate

Quality Metric	Description	Target
Silhouette Score	Measures cluster cohesion and separation	Closer to 1.0 is better
Davies-Bouldin Index	Average similarity between clusters	Closer to 0.0 is better

Module D — NEPRA Billing Engine (billing_engine.py)

The billing engine implements Pakistan's official NEPRA electricity tariff structure, computing bills across five consumption slabs with a fixed base charge and 17% federal tax.

Tariff Slab	Rate (PKR/unit)	Category	Notes
0 – 100 kWh	PKR 18.50	Protected (Low-Income)	Subsidized rate, protected status
101 – 200 kWh	PKR 20.50	Normal	Standard residential rate
201 – 300 kWh	PKR 24.75	High	Upper-middle consumption
301 – 500 kWh	PKR 28.90	Very High	Commercial/AC-heavy usage
500+ kWh	PKR 32.50	Industrial	Maximum rate tier

Bill Formula:

Total Bill = Base Charge (PKR 300) + (Units × Rate/Unit) + Tax (17%)

Module E — Agentic AI Recommendations (generative_ai.py)

- LLM Provider: Groq API (free tier) using Mixtral-8x7b-32768 model
- Technique: RAG-based prompt engineering — tariff rules injected into system context
- Output: Personalized JSON-structured energy-saving strategies with PKR savings estimates
- Fallback: Rule-based recommendations generated automatically if API key is unavailable
- Configuration: Temperature=0.7, Max Tokens=500, customizable via config.py

7. Tech Stack

Component	Technology	Version / Notes
Language	Python 3	v3.8+ recommended
Dashboard / UI	Streamlit	>=1.28.0, dark mode, 3-tab layout
ML Models	Scikit-learn, XGBoost	Random Forest + Gradient Boosting
Time-Series	Facebook Prophet	Seasonal forecasting (disabled on cloud)
Data Processing	Pandas, NumPy, SciPy	Feature engineering and statistics
Clustering & PCA	Scikit-learn KMeans + PCA	Consumer segmentation, 2D visualization
Visualization	Plotly, Seaborn, Matplotlib	Interactive charts and heatmaps
LLM / AI Advisory	Groq API (Mixtral-8x7b)	RAG-based prescriptive recommendations
Statistical Decomp.	Statsmodels	Trend + Seasonality + Residual split
Model Persistence	Joblib	Save/load trained model files
Configuration	config.py	Central hyperparameter and constant store

8. Project Structure

File	Responsibility
app.py	Main Streamlit dashboard — entry point, 3-tab layout, KPI cards, sliders
config.py	All constants: NEPRA slabs, ML hyperparameters, theme colors, LLM settings
data_engineering.py	DataPreprocessor class: imputation, outlier removal, feature extraction, decomposition
models.py	PredictiveModels + ModelComparison classes: RF, XGBoost training and evaluation
clustering.py	ConsumerSegmentation class: K-Means, PCA, silhouette score, segment profiles
billing_engine.py	BillingEngine class: NEPRA tariff calculation, savings analysis, peak/off-peak comparison
generative_ai.py	AIOptimizer class: Groq LLM integration, RAG prompts, fallback recommendations
visualizations.py	Visualizations class: Plotly heatmaps, decomposition plots, scatter plots, pie charts

File	Responsibility
household_power_consumption_50k.txt	Real dataset: 50,000 records of household power consumption (semicolon-delimited)
requirements.txt	Python dependencies for pip installation
README.md	Setup instructions, controls guide, algorithm explanations

9. Dashboard & GUI Features

Tab 1 — Dashboard & Live Predictor

- KPI cards: Average monthly consumption, average bill (PKR), protected-status user percentage, dataset size
- Interactive sliders: Adjust monthly consumption (30–500 kWh) and temperature (10–50 °C) in real time
- Predict Next Month's Bill button: Triggers NEPRA calculation, displays slab, rate, status, and total bill
- Bill Breakdown: Side-by-side Plotly pie chart and formatted data table (Base / Units / Tax components)

Tab 2 — Core AI Engine Analytics

Sub-Tab	Content
EDA Analytics Hub	Plotly correlation heatmap across all numeric columns; seasonal decomposition (trend/seasonality/residual)
Predictive Models	Side-by-side RF vs XGBoost metric table; best model highlighted; actual vs predicted chart with 3-month forecast shading
Consumer Segmentation	2D PCA scatter plot with 4 colour-coded clusters; segment profile statistics table
Feature Analysis	Consumption and temperature descriptive statistics; consumption-by-season box plots

Tab 3 — AI Bill Optimizer

- User inputs: monthly consumption (number field) and consumer segment (dropdown)
- Savings Opportunities: Three metric cards showing units to cut for Protected Status, Lower Slab, and Off-Peak Shifting savings
- AI Advisory: Groq LLM-generated personalized text recommendations displayed in info block
- Action Plan: Structured 4-step plan (Immediate / Short-term / Medium-term / Long-term) with estimated PKR savings

Theme & Design

- Dark mode background: #0e1117 (Streamlit default dark)
- Accent colors: Electric Blue (#00D9FF) and Neon Green (#39FF14) for headings and metrics
- Metric cards use gradient backgrounds with left-border accent styling

10. Testing

- Verified BillingEngine produces correct PKR amounts across all five NEPRA tariff slabs
- Confirmed protected-status threshold correctly flags consumption ≤ 100 kWh
- Tested DataPreprocessor handles missing values and outliers without dropping valid seasonal rows
- Validated K-Means clustering assigns all records to exactly one of four segments
- Confirmed PCA reduces features to 2 components with cumulative variance reported correctly
- Tested ModelComparison selects the model with the highest R^2 as the best model
- Verified actual vs predicted chart correctly displays the 3-month forecast shading window
- Tested LLM fallback logic: recommendations are generated even when no API key is set
- Confirmed dataset falls back to synthetic 36-month data when the .txt file is unavailable
- Validated savings calculation: units-to-cut and PKR savings are accurate across all slab transitions

11. Results

- XGBoost outperformed Random Forest on R^2 in most test scenarios, though both achieved high accuracy
- Random Forest provided more interpretable feature importance scores, confirming temperature and month as top predictors
- Prophet model architecture is implemented and available locally; disabled for Streamlit Cloud compatibility
- K-Means segmentation produced well-separated consumer clusters with positive Silhouette Scores
- NEPRA billing engine correctly computed bills and savings across all five tariff slabs in every tested scenario
- LLM advisory module generated context-aware, slab-specific recommendations for all four consumer segments
- Dashboard loaded and rendered all three tabs without errors on 50,000-row real dataset
- Off-peak shifting analysis demonstrated potential monthly savings for consumers in the 301–500 kWh slab

12. Conclusion

This project demonstrates how a comprehensive AI/ML pipeline can be applied to address a pressing real-world challenge in Pakistan — electricity bill management and energy optimization. By combining supervised learning, unsupervised clustering, time-series forecasting, rule-based billing, and generative AI advisory, the system delivers a full-spectrum solution that is both technically rigorous and practically useful.

The modular architecture — spanning `data_engineering.py`, `models.py`, `clustering.py`, `billing_engine.py`, `generative_ai.py`, and `visualizations.py` — ensures clarity, maintainability, and extensibility. The Streamlit dashboard bridges the gap between complex AI outputs and end-user accessibility, presenting predictions, segmentation results, and personalized recommendations through an intuitive, visually engaging interface.

The project reinforces core AI principles including end-to-end ML pipeline design, the trade-offs between model accuracy and interpretability, the value of domain knowledge (NEPRA tariffs) in system design, and the emerging role of LLMs as prescriptive AI agents. Overall, the Pakistan Electricity Smart Bill Predictor is a strong example of AI applied with purpose — combining technical depth with real-world impact.